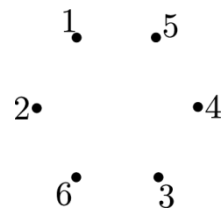
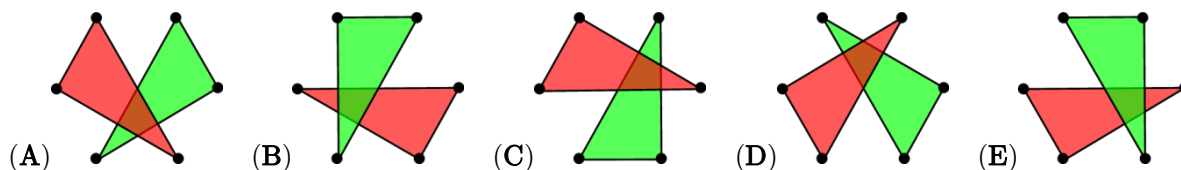


3 points

1. Six points are numbered as shown. Kirsten draws two triangles, one by joining the even numbered points and one by joining the odd numbered points, and colours the inside of one of the triangles red and the inside of the other green. Which of the five options shows the picture Kirsten draws?

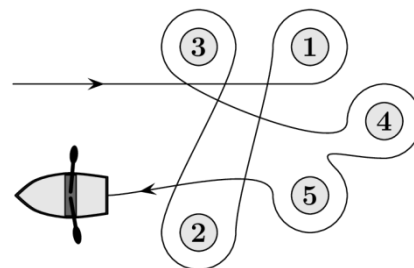


六个点的编号如右图所示。克里斯将编号都是偶数的点连接起来，将编号都是奇数的点连接起来，构成两个三角形，并将其中一个三角形涂成红色，另一个涂成绿色。请问下面哪个选项是克里斯画的图？



2. Eileen rowed around five buoys, as shown. Which buoys did Eileen row around in an anticlockwise direction?

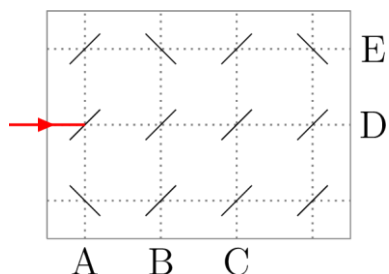
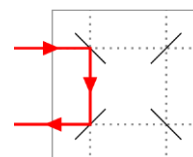
艾琳绕着五个浮标划船，如图所示。请问她绕着哪几个浮标划船时是逆时针方向？



- (A) 1 and 4 1 和 4 (B) 2, 3 and 5 2、3 和 5
(C) 2 and 3 2 和 3 (D) 1, 4 and 5 1、4 和 5
(E) 1 and 3 1 和 3

3. Laser beams reflect in mirrors in the way shown in the picture. At which letter will this laser beam end?

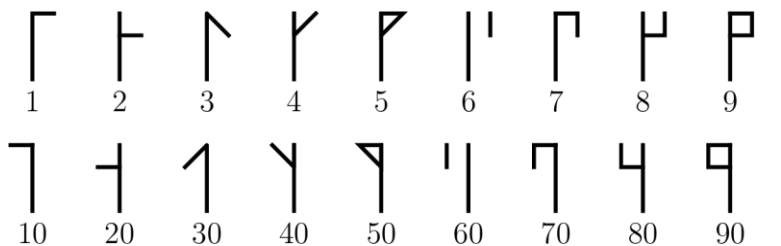
激光束在镜子中的反射情况如右图所示。那么下图中的激光束将从哪个字母的位置反射出来？



- (A) A (B) B (C) C (D) D (E) E

4. Cistercian numerals were used in the early thirteenth century. Any integer from 1 to 99 can be represented by a single glyph formed by combining two of the glyphs shown below.

西多会数字系统在 13 世纪早期开始被使用。从 1 到 99 的任何整数都可以用下图所示的字形单独或组合表示。



The glyph for 24 looks like , the glyph for 81 looks like .

数字 24 的字形是 , 数字 81 的字形是 .

and the glyph for 93 looks like .

数字 93 的字形是 .

What does the glyph for 45 look like?

下面哪个是数字 45 的字形?

- (A)  (B)  (C)  (D)  (E) 

5. Marbles are sold in packages of 5, 10 or 25. Tom buys exactly 95 marbles. What is the minimum number of packages he could buy?

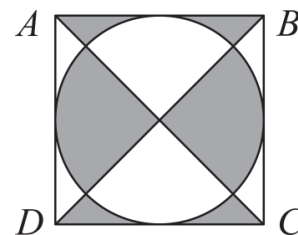
弹珠以 5 颗、10 颗或 25 颗为一包出售。汤姆最少需要买几包可以正好买到 95 颗弹珠?

- (A) 4 (B) 5 (C) 7 (D) 8 (E) 10

6. ABCD is a square with side-length 10 cm. What is the area of the shaded part?

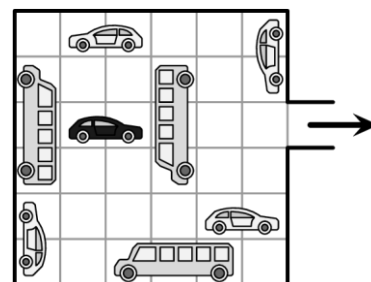
正方形 ABCD 的边长为 10 cm。请问阴影部分的面积是多少?

- (A) 40 cm^2 (B) 45 cm^2 (C) 50 cm^2
(D) 55 cm^2 (E) 60 cm^2



7. In the garage shown in the picture, vehicles can only move forward or backward but cannot turn. What is the smallest number of vehicles that have to move for the black car to be able to exit the garage?

在图中所示的车库中，所有车辆只能前进或后退，不能转弯。为了让黑色汽车能够驶离车库，最少需要移动多少辆汽车？

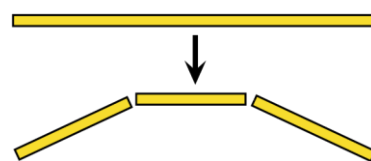


- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

8. Giulia has one long string of spaghetti she needs to make smaller. Every time she breaks one piece of spaghetti, it becomes three pieces as shown in the picture. Which of the following numbers of pieces could she not get?

朱莉娅有一长串意大利面条，她需要把它变短。

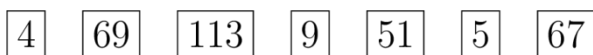
她每折断一次面条，面条就会变成三截，如图所示。请问朱莉娅不可能得到以下哪种数量的面条？



- (A) 13 (B) 17 (C) 20 (D) 23 (E) 25

9. Bodil rearranges the 7 pieces shown to get the smallest possible 12 digit number.

小博将下图所示的 7 张数字卡片重新排列后，得到一个尽可能小的 12 位数。



What are the last 3 digits of this number?

请问这个数的最后 3 位数字是多少？

- (A) 699 (B) 113 (C) 551 (D) 967 (E) 459

10. Which of the following fractions of a whole turn should the Ferris wheel turn to bring a white pod to the top?

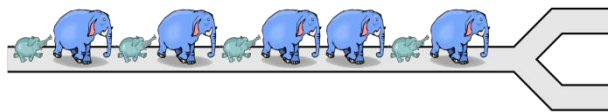
摩天轮要转动整圈的几分之几才能将一个白色吊舱转到顶部？



- (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{6}$ (D) $\frac{1}{12}$ (E) $\frac{5}{6}$

4 points

11. Five big elephants and four small ones are walking along a path, as shown. When they reach the junction, each elephant turns either to the left or to the right.

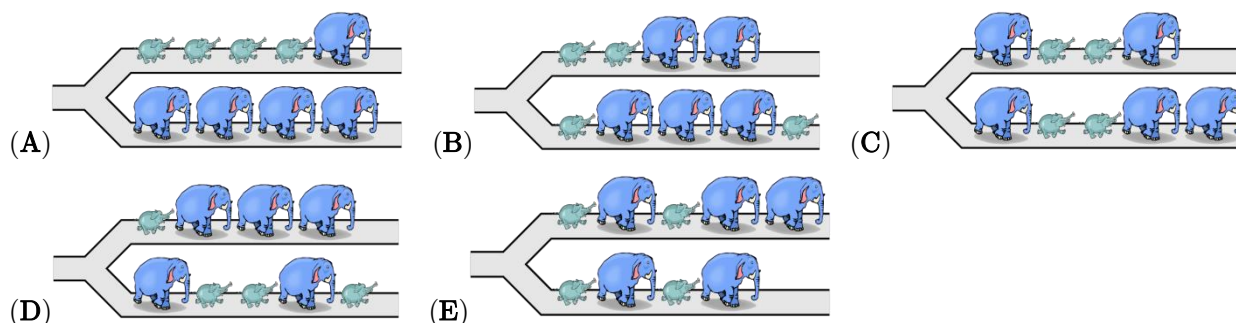


Which of the following cannot be the situation after they all pass the junction?

如图所示，五头大象和四头小象正沿着一条小路行走。

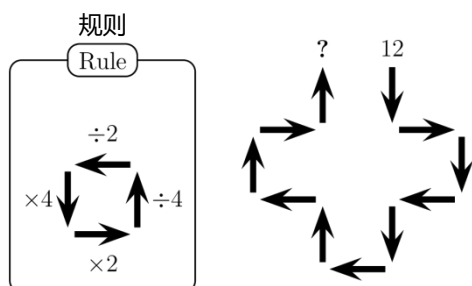
到达岔路口时，每头象要么向左走，要么向右走。

在它们全部通过岔路口后，不可能出现以下哪种情况？



12. Clara starts with 12 and follows the arrows using the rules shown in the picture on the left.

拉拉从数字 12 开始，沿着箭头的方向按照左图所示的规则前进。

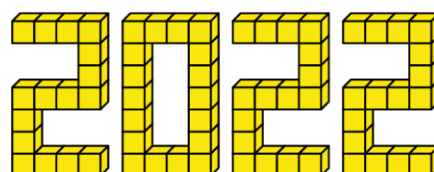


What number will she finish with?

请问她最后会得到哪个数字？

- (A) 3 (B) 6 (C) 12 (D) 24 (E) 48

13. Masoud and his friends built the number 2022 with 66 cubes, as shown in the picture. They painted the whole surface of the structure yellow. How many of the cubes have exactly 4 faces painted?



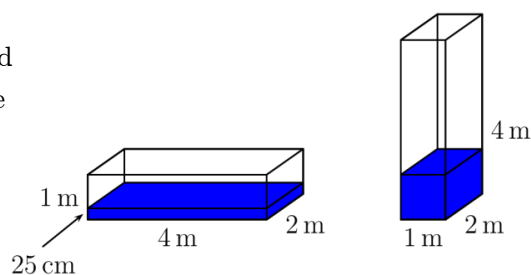
小马和他的朋友们用 66 个小立方体搭建了数字 2022，如图所示。他们把整个结构的表面涂成黄色。请问有多少个小立方体恰好有 4 个面都被涂上了颜色？

- (A) 16 (B) 30 (C) 46 (D) 54 (E) 60

14. A rectangular-based water tank has dimensions $1\text{ m} \times 2\text{ m} \times 4\text{ m}$.

It contains water to a depth of 25 cm, as shown in the left-hand picture. The tank is turned so that a $1\text{ m} \times 2\text{ m}$ face becomes the base, as shown in the right-hand picture. What is the depth of the water now?

一个底面是长方形的水箱尺寸为 $1\text{ m} \times 2\text{ m} \times 4\text{ m}$ 。如左图所示，它的水深为 25 cm。将水箱翻转，使 $1\text{ m} \times 2\text{ m}$ 的面作为底面，如右图所示。请问现在的水深是多少？

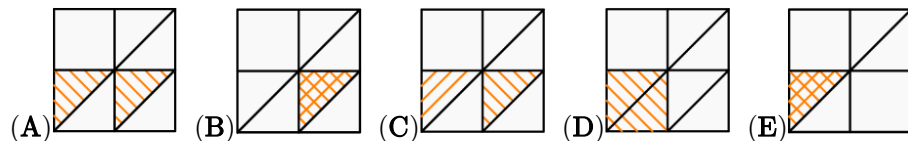
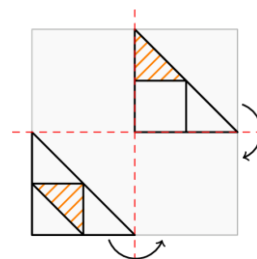


- (A) 25 cm (B) 50 cm (C) 75 cm (D) 1 m (E) 1.25 m

15. The picture shows a piece of transparent paper with a design drawn on it.

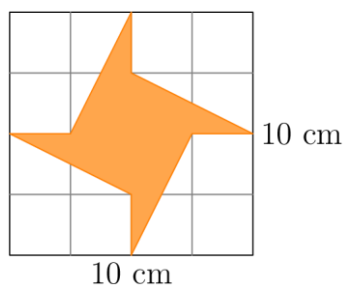
The paper is then folded twice, as shown. What would be seen on the folded paper?

一张透明纸上画有右图所示的图案，将这张纸按照箭头方向折叠两次。请问折叠后的纸上会出现下面哪种图案？



16. The area of the square is 100 cm^2 .

大正方形的面积是 100 cm^2 。



What is the area of the shaded figure?

请问阴影部分的面积是多少？

- (A) 20 cm^2 (B) 25 cm^2 (C) 30 cm^2 (D) 35 cm^2 (E) 40 cm^2

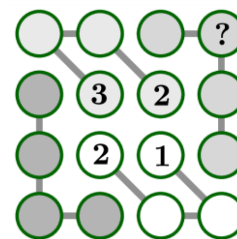
17. The year 2022 is a special year because the digit 2 appears three times. This is the third time Eva the tortoise has lived through such a year with three identical digits. What is the youngest Eva could be by the end of 2022?

2022 年是一个特殊的年份，因为数字 2 出现了 3 次。对于乌龟伊娃来说，这是它第三次经历这样一个有三个相同数字的年份。请问 2022 年结束时，伊娃最少多少岁？

- (A) 18 (B) 20 (C) 22 (D) 23 (E) 134

18. Andrew wants to complete the picture so that each row, each column and each set of four circles connected by line segments contains the four numbers 1, 2, 3, and 4. What number should he write in the circle containing the question mark?

安安想完成这幅图，使每一行、每一列和每一组由线段连接的四个圆圈都包含数字 1、2、3、4。请问他应该在问号的圆圈里填哪个数字？



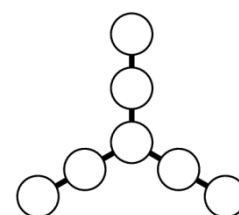
- (A) 1 (B) 2 (C) 3 (D) 4 (E) It cannot be determined 无法确定

19. Lisa has 4 dogs. Each of the 4 dogs weighs an integer number of kg. No two of them weigh the same. Their total weight is 60 kg. The second heaviest dog weighs 28 kg. How heavy is the third heaviest dog?
丽丽有 4 只狗，每只狗的体重都是整数 kg，且重量各不相同。已知它们的总重量为 60 kg，第二重的狗有 28 kg。请问第三重的狗体重是多少？

- (A) 2 kg (B) 3 kg (C) 4 kg (D) 5 kg (E) 6 kg

20. Jessi writes the seven numbers 3, 4, 5, 6, 7, 8, and 9 in the circles in the picture so that the sums of the three numbers on each line are equal. What is the largest possible sum of three numbers on a line that Jessi can get?

杰西将 3、4、5、6、7、8、9 这七个数写在图中的圆圈里，使得每条线上的三个数字之和都相等。请问杰西能得到的每条线上的三个数字之和的最大可能值是多少？



- (A) 28 (B) 18 (C) 22 (D) 16 (E) 20

5 points

21. Some glasses are stacked on top of each other. A stack of 8 glasses is 42 cm high and a stack of 2 glasses is 18 cm high. How high is a stack of 6 glasses?

一些玻璃杯叠放在一起。一摞 8 个玻璃杯有 42 cm 高，一摞 2 个玻璃杯有 18 cm 高。请问一摞 6 个玻璃杯有多高？



- (A) 22 cm (B) 24 cm (C) 28 cm (D) 34 cm (E) 40 cm

22. In the picture below, each animal represents a positive integer and different animals represent different integers. The sum of the two integers in each column is written below that column. What is the largest possible sum of the four integers in the first row?

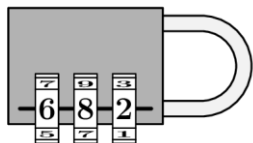
在下图中，每只动物代表一个正整数，不同的动物代表不同的整数。每一列下方的数字表示该列两个整数之和。请问第一行中四个整数之和的最大可能值是多少？

				?
				
15	11	3	7	

- (A) 18 (B) 19 (C) 20 (D) 21 (E) 22

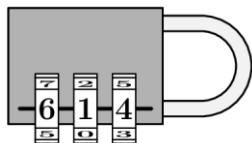
23. To unlock this lock, you get the following four hints.

使用以下四个提示来解开这把锁。



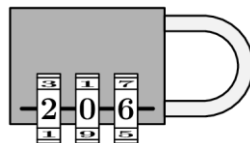
One of these digits is correct and in the right place.

其中有一个数字正确且在正确的位置上



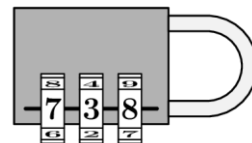
One of these digits is correct but in the wrong place.

其中有一个数字正确但不在正确的位置上



Two of these digits are correct but in the wrong place.

其中有两个数字正确但不在正确的位置上



All of these digits are incorrect.

所有数字都不正确

What is the correct code for the lock?

请问这把锁的正确密码是什么？

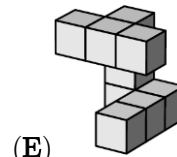
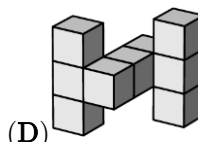
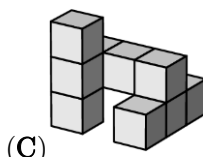
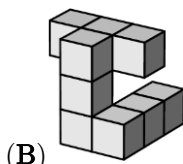
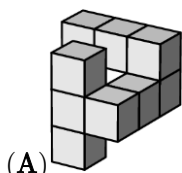
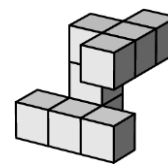
- (A) 604 (B) 082 (C) 640 (D) 042 (E) 046

24. Anna has the shape shown on the right.

Which of the following shapes is the same as Anna's?

安娜有一个如右图所示的积木。

以下哪个积木的形状与安娜的积木形状相同？



25. Werner chooses four of the numbers 2, 3, 4, 5 and 6 and writes

one in each box so that the calculation is correct. How many of the five numbers could Werner write in the shaded box?

小伟从 2、3、4、5 和 6 中选出四个数字填在右边每个框中，

使得算式成立。请问在这五个数字中，有几个数字可以填在阴影框中？

$$\square + \square - \square = \square$$

(A) 1

(B) 2

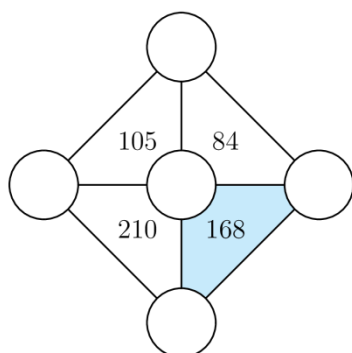
(C) 3

(D) 4

(E) 5

26. The numbers 3, 4, 5, 6 and 7 are to be placed in the five circles below so that the number inside each triangle is the product of the three numbers on its vertices.

将数字 3、4、5、6 和 7 放置在下面的五个圆圈中，使得每个三角形内部的数字是其顶点上三个数字的乘积。



What is the sum of the three numbers on the vertices of the coloured triangle?

请问涂色三角形的顶点上的三个数字之和是多少？

(A) 12

(B) 14

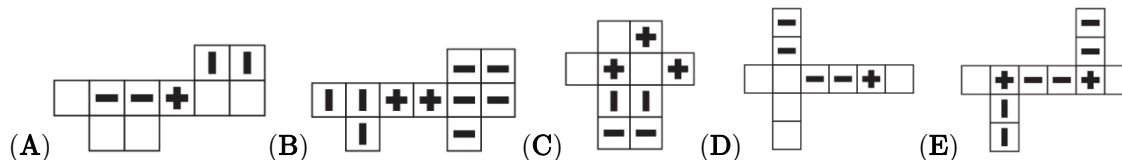
(C) 15

(D) 17

(E) 18

27. Which of the following nets cannot be folded into the solid 

下面哪个展开图不能折叠成这个立体形状  ?



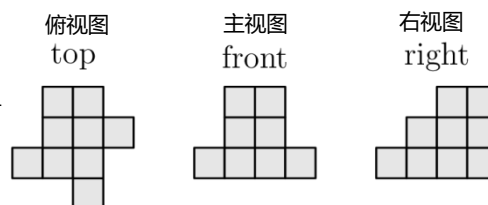
28. The four villages A, B, C and D lie along a road in that order. The distance between neighbouring villages is 10 km. There are 10 students who live in village A, 20 students who live in village B, 30 students who live in village C and 40 students who live in village D. The villagers want to build a school so that the total distance travelled by the students when going to school is as small as possible. Where should they build the school?

四个村庄 A、B、C、D 依次排列在一条道路上。相邻村庄之间的距离为 10 km。A 村有 10 名学生，B 村有 20 名学生，C 村有 30 名学生，D 村有 40 名学生。现在村民想建一所学校，使得这些学生去学校的总路程尽可能小。请问村民应该在哪里建学校？

- (A) in A A 村 (B) in B B 村 (C) in the middle between B and C B 村和 C 村中间
(D) in C C 村 (E) in D D 村

29. The three pictures show a structure made from cubes as seen from the top, from the front and from the right. What is the maximum number of cubes that could have been used to build the structure?

一个大立方体由若干个小立方体组成，它的俯视图、主视图和右视图如图所示。请问这个大立方体最多由多少个小立方体组成？



- (A) 18 (B) 19 (C) 20 (D) 21 (E) 22

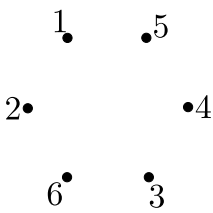
30. 30 people are sitting round a circular table. Some of them are wearing a hat. Those who do wear a hat always tell the truth while those who do not wear a hat can either lie or tell the truth. Each person says "At least one of my two neighbours is not wearing a hat." What is the largest number of people who could be wearing a hat?

30 个人围坐在一张圆桌旁。他们中的一些人戴着帽子。戴帽子的人总是说真话，不戴帽子的人可以说假话也可以说真话。但每个人都说：“我左右相邻的两个人当中至少有一个人没戴帽子。”请问最多有多少个戴帽子的人？

- (A) 5 (B) 10 (C) 15 (D) 20 (E) 25

3 points

1 (MX). Six points are numbered as shown. Kirsten draws two triangles, one by joining the even numbered points and one by joining the odd numbered points, and colours the inside of one of the triangles red and the inside of the other green. Which of the five options shows the picture Kirsten draws?

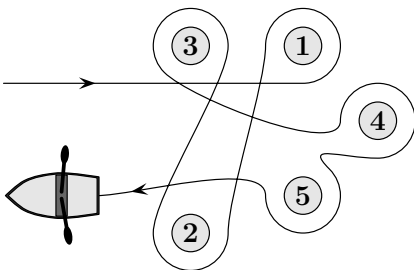


- (A)
- (B)
- (C)
- (D)
- (E)

SOLUTION: 1, 3 and 5 are odd numbers; 2, 4 and 6 are even numbers. Connecting these three points you get the drawing E.

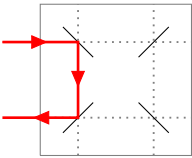
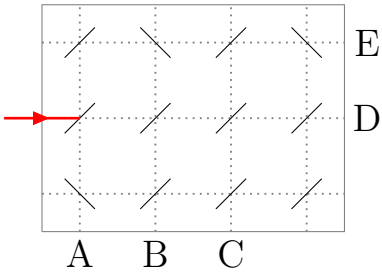
2 (DE). Eileen rowed around five buoys, as shown. Which buoys did Eileen row around in an anticlockwise direction?

- (A) 1 and 4
(B) 2, 3 and 5
(C) 2 and 3
(D) 1, 4 and 5
(E) 1 and 3

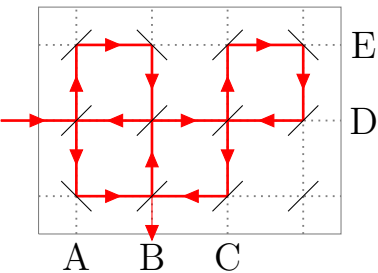


SOLUTION: When Eileen turns around a buoy in counterclockwise direction, the buoy is at her left. This is the case only with buoys 1 and 3, so the solution is E.

3 (DK). Laser beams reflect in mirrors in the way shown in the picture. At which letter will this laser beam end?

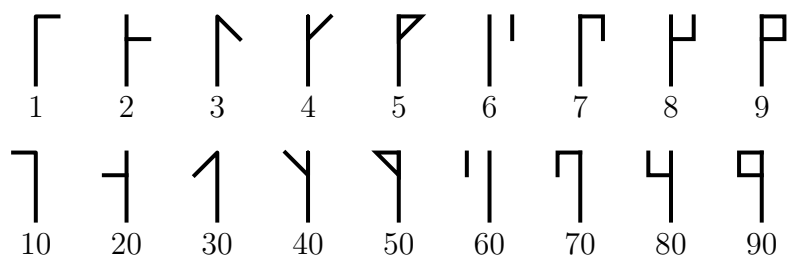


- (A) A
(B) B
(C) C
(D) D
(E) E



SOLUTION: It reaches the letter B.

4 (UZ). Cistercian numerals were used in the early thirteenth century. Any integer from 1 to 99 can be represented by a single glyph formed by combining two of the glyphs shown below.



The glyph for 24 looks like , the glyph for 81 looks like

and the glyph for 93 looks like

What does the glyph for 45 look like?

- (A) (B) (C) (D) (E)

SOLUTION: The number $45 = 40 + 5$, so you combine the two glyphs which stand for 40 and 5. This is the glyph D.

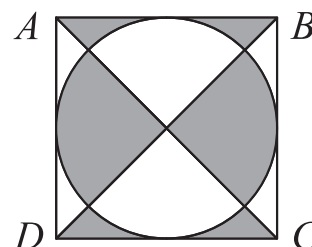
5 (IQ). Marbles are sold in packages of 5, 10 or 25. Tom buys exactly 95 marbles. What is the minimum number of packages he could buy?

- (A) 4 (B) 5 (C) 7 (D) 8 (E) 10

SOLUTION: In order to get the smallest number of bags, Tom has to start with the bags that hold the most marbles. $95 = 25 + 25 + 25 + 10 + 10$, so Tom needs to buy at least 5 packages.

6 (RU). ABCD is a square with side-length 10 cm. What is the area of the shaded part?

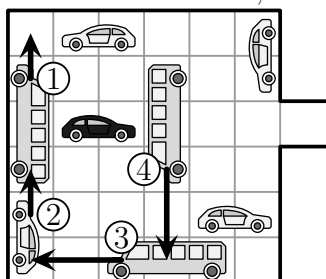
- (A) 40 cm^2 (B) 45 cm^2
 (C) 50 cm^2 (D) 55 cm^2
 (E) 60 cm^2



SOLUTION: If you fold the figure along one of the diagonals you can notice that the white parts of the figure exactly cover the shaded ones. So we conclude both have the same area, which is half of the square. The area of the square $ABCD$ is $10\text{cm} \cdot 10\text{cm} = 100\text{cm}^2$. Half of this is 50cm^2 , so the correct is C.

A 5x5 grid world environment. The robot is at (1, 1). Obstacles are at (1, 3), (1, 4), (2, 3), (2, 4), (3, 3), (3, 4), (4, 3), (4, 4), (5, 3), and (5, 4). The goal is at (5, 5). The grid is shown with a black border and a large arrow pointing to the right.

- SOLUTION: In total, four vehicles had to move, so the correct answer is C.



A diagram illustrating the deflection of a beam under a point load. A horizontal yellow bar represents the beam. A black arrow points downwards from the center of the beam, indicating the direction of the applied load. Below the beam, the deflected shape is shown as a yellow bar that has bent downwards at the point of application of the load, forming a V-shape. The original straight position of the beam is indicated by a dashed line above the deflected shape.

- SOLUTION:** Solution: After each break the number of pieces increases by 2, since 2 new pieces are made. Since there is one piece at the beginning, and Giulia always adds an even number of pieces in each break, the number of pieces will always be odd. Therefore it is impossible for an even number of pieces to remain, which means that the answer is (C) 20.

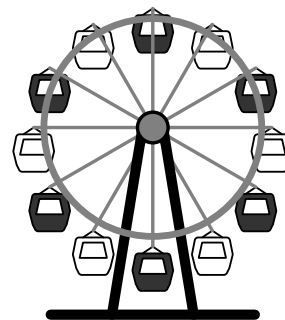
4 69 113 9 51 5 67

(A) 699 (B) 113 (C) 551 (D) 967 (E) 459

3

10 (NO). Which of the following fractions of a whole turn should the Ferris wheel turn to bring a white pod to the top?

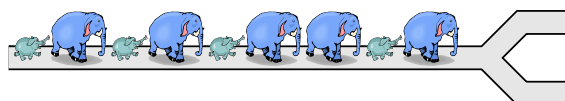
- (A) $\frac{1}{2}$
- (B) $\frac{1}{3}$
- (C) $\frac{1}{6}$
- (D) $\frac{1}{12}$
- (E) $\frac{5}{6}$



SOLUTION: Solution: There are 12 wagons on the ferris wheel. So, with a rotation of a $\frac{1}{12}$ of a whole turn, all the wagons moved by one position, so a white wagon will be on top, thich is answer D.

4 points

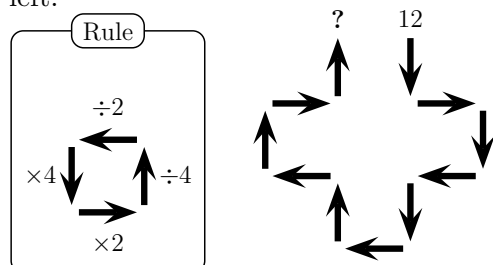
11 (GR). Five big elephants and four small ones are walking along a path, as shown. When they reach the junction, each elephant turns either to the left or to the right. Which of the following cannot be the situation after they all pass the junction?



- (A)
- (B)
- (C)
- (D)
- (E)

SOLUTION: The last elephant originally is a small one, so a small elephant must be the last one on at least one branch after the junction. This leaves out option C. Note that all others are possible.

12 (PH). Clara starts with 12 and follows the arrows using the rules shown in the picture on the left.

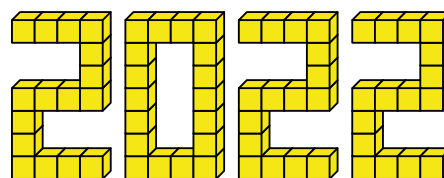


What number will she finish with?

- (A) 3
- (B) 6
- (C) 12
- (D) 24
- (E) 48

SOLUTION: There are 3 down arrows, 2 right arrows, 3 left arrows, and 3 up arrows. Therefore, the overall effect is just one left arrow, which is division by 2. So the final result is $B = 12 \div 2 = 6$.

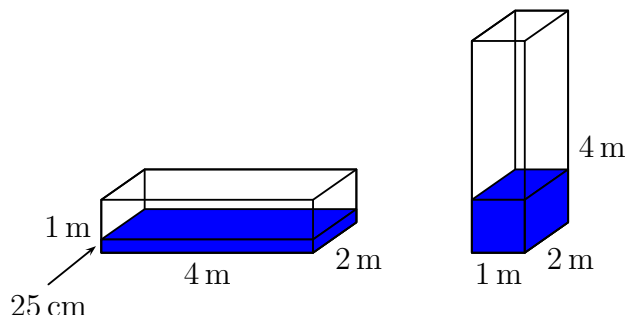
13 (IR). Masoud and his friends built the number 2022 with 66 cubes, as shown in the picture. They painted the whole surface of the structure yellow. How many of the cubes have exactly 4 faces painted?



- (A) 16 (B) 30 (C) 46 (D) 54 (E) 60

SOLUTION: Each number 2 has two cubes with more than 4 colored faces (in the ends). The number 0 has no cubes with more than 4 colored faces. Therefore the answer is $66 - 3 \times 2 = 60$.

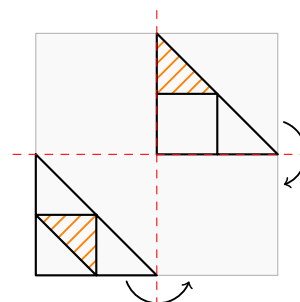
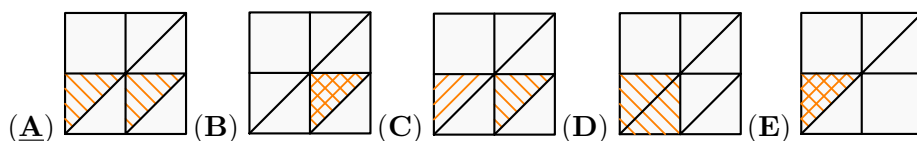
14 (GR). A rectangular-based water tank has dimensions 1m x 2m x 4m. It contains water to a depth of 25 cm, as shown in the left-hand picture. The tank is turned so that a 1m x 2m face becomes the base, as shown in the right-hand picture. What is the depth of the water now?



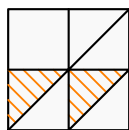
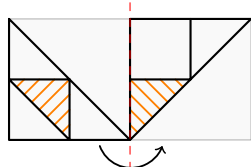
- (A) 25 cm (B) 50 cm (C) 75 cm
(D) 1 m (E) 1.25 m

SOLUTION: Originally the height of the water is $\frac{1}{4}$ of the height of the deposite, so it occupies $\frac{1}{4}$ of its volume. When it is turned, the water will of course still occupy $\frac{1}{4}$ of its volume, so it level will be 1 m (which is $\frac{1}{4}$ of 4m).

15 (IR). The picture shows a piece of transparent paper with a design drawn on it. The paper is then folded twice, as shown. What would be seen on the folded paper?

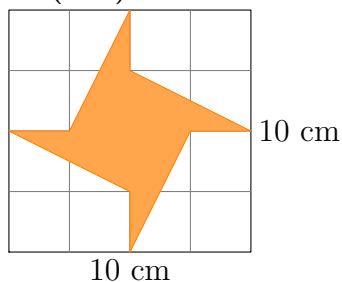


SOLUTION: After the first folding it looks like this:



And after the second like this:

16 (DK). The area of the square is 100 cm^2 .



What is the area of the shaded figure?

- (A) 20 cm^2 (B) 25 cm^2 (C) 30 cm^2 (D) 35 cm^2 (E) 40 cm^2

SOLUTION: If we draw a vertical and horizontal line along the middle lines of the square, we realize that the figure is formed by 4 right triangles each of base $\frac{10}{4}$ and height $\frac{10}{2}$. Together they form a rectangle of base $\frac{10}{4}$ and height 10, whose area is $\frac{100}{4} = 25$.

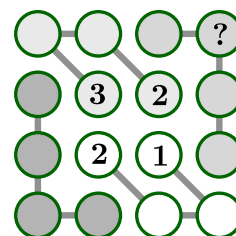
17 (AT). The year 2022 is a special year because the digit 2 appears three times. This is the third time Eva the tortoise has lived through such a year with three identical digits. What is the youngest Eva could be by the end of 2022?

- (A) 18 (B) 20 (C) 22 (D) 23 (E) 134

SOLUTION: There were three of the same digits in 2022, 2000, and 1999, so Eva will be at least 23 this year.

18 (CT). Andrew wants to complete the picture so that each row, each column and each set of four circles connected by line segments contains the four numbers 1, 2, 3, and 4. What number should he write in the circle containing the question mark?

- (A) 1 (B) 2 (C) 3 (D) 4
(E) It cannot be determined



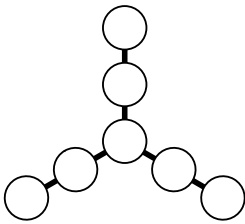
SOLUTION: At the first row, the first two circles from left to right are either 1 – 4 or 4 – 1. Therefore the the third circle from left to right has to be 2 or 3, but since there is a 2 at the third column, left to right, we are forced to put the number 2 in the last circle, left to right.

19 (GR). Lisa has 4 dogs. Each of the 4 dogs weighs an integer number of kg. No two of them weigh the same. Their total weight is 60 kg. The second heaviest dog weighs 28 kg. How heavy is the third heaviest dog?

- (A) 2 kg (B) 3 kg (C) 4 kg (D) 5 kg (E) 6 kg

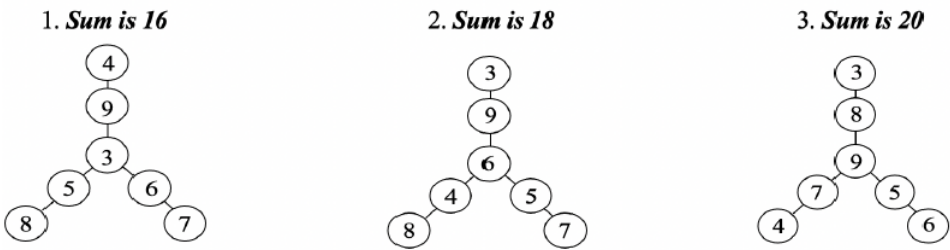
SOLUTION: If the third heaviest was 3 or more kilos then the total weight would be at least $29 + 28 + 3 + 1 = 61 > 60$. So the third heaviest is at most 2 and hence 2, leaving 1 for the lightest. If you wish an example, the unique solution is $29 + 28 + 2 + 1 = 60$. Another solution: The heaviest could not be 30 or more kilos because their total weight would then be at least $30 + 28 + 2 + 1 = 61 > 60$. So the heaviest is 29 kilos. The two lightest would then weigh $60 - 29 - 28 = 3$ kilos, so the two smallest ones can only be 2 + 1 kilos.

20 (PH). Jessi writes the seven numbers 3, 4, 5, 6, 7, 8, and 9 in the circles in the picture so that the sums of the three numbers on each line are equal. What is the largest possible sum of three numbers on a line that Jessi can get?



- (A) 28
- (B) 18
- (C) 22
- (D) 16
- (E) 20

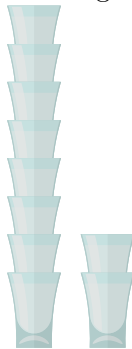
SOLUTION: There are three possible answers as shown below. The greatest possible sum is 20 .



Solution 2. The sum would be greatest if we could put 9 in the middle circle. We can see that $8 + 3 = 7 + 4 = 6 + 5 = 11$, therefore the greatest possible sum is $11 + 9 = 20$.

5 points

21 (NO). Some glasses are stacked on top of each other. A stack of 8 glasses is 42 cm high and a



stack of 2 glasses is 18 cm high. How high is a stack of 6 glasses?

- (A) 22 cm
- (B) 24 cm
- (C) 28 cm
- (D) 34 cm
- (E) 40 cm

SOLUTION: $42 - 18 = 24$ is the difference between the stack of 8 and 2 glasses. Hence the increase for one extra glass is $\frac{24}{6} = 4$. We can add this four times to the stack of 2 glasses. $18 + 16 = 34cm$.

22 (HK). In the picture below, each animal represents a positive integer and different animals represent different integers. The sum of the two integers in each column is written below that column. What is the largest possible sum of the four integers in the first row?

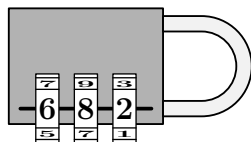
				?
				
15	11	3	7	

- (A) 18
- (B) 19
- (C) 20
- (D) 21
- (E) 22

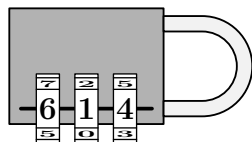
SOLUTION: In the third column, two different non-zero numbers add up to 3, so the only possibility is $3 = 1 + 2$. In the fourth column, two different non-zero numbers add up to 7. Given numbers 1 and 2 already appear in the third column, so the only possibility here is $7 = 3 + 4$. Similarly for the two numbers in the second column, we must have $11 = 5 + 6$, and for the two numbers in the first column

they must be 7 and 8 to form $7 + 8 = 15$. To make the sum of the four numbers in the first row as large as possible, we can make the larger number of each column to be in the first row, which are 8, 6, 2 and 4 respectively. The answer is $8 + 6 + 2 + 4 = 20$.

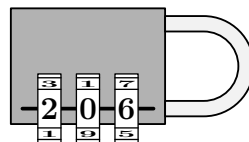
23 (IR). To unlock this lock, you get the following four hints.



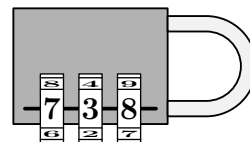
One of these digits is correct and in the right place.



One of these digits is correct but in the wrong place.



Two of these digits are correct but in the wrong place.



All of these digits are incorrect.

What is the correct code for the lock?

- (A) 604 (B) 082 (C) 640 (D) 042 (E) 046

SOLUTION: Locker 738: All these numbers are not used.

Lockers 682 and 614: We cannot choose 6. Hence, 2 is on the right position in the code.

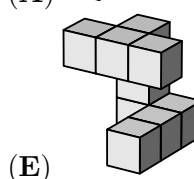
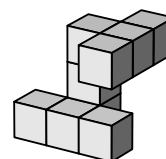
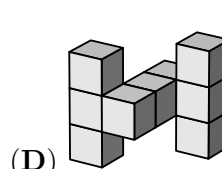
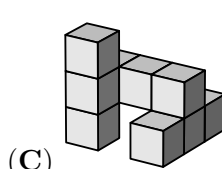
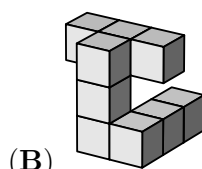
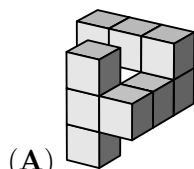
Locker 206: 0 is on the first position in the code.

Locker 614: 4 is on the second position in the code.

The whole code is 042.

24 (CT). Anna has the shape shown on the right.

Which of the following shapes is the same as Anna's?



SOLUTION: No solution needed, look at the pictures and turn them.

Each figure is made up of three parts ("towers") consisting of three small cubes glued one after the other. We will see the answer when we look at the places where these "towers" touch each other.

25 (UK). Werner chooses four of the numbers 2, 3, 4, 5 and 6 and writes one in each box so that the calculation is correct.

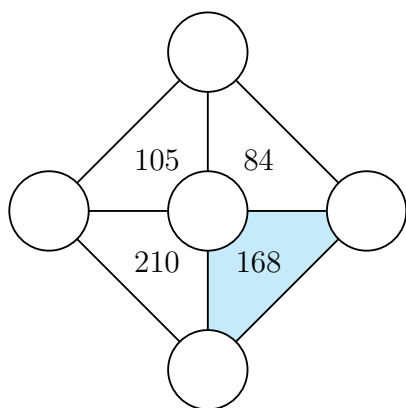
$$\square + \square - \square = \square$$

How many of the five numbers could Werner write in the shaded box?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION: You can get all 5 numbers as the result on the right, for example this way. $3 + 5 - 6 = 2$, $2 + 6 - 5 = 3$, $2 + 5 - 3 = 4$, $2 + 6 - 3 = 5$, $3 + 5 - 2 = 6$

26 (IR). The numbers 3, 4, 5, 6 and 7 are to be placed in the five circles below so that the number inside each triangle is the product of the three numbers on its vertices.

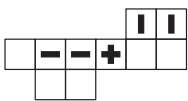
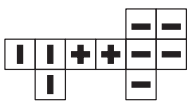
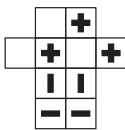
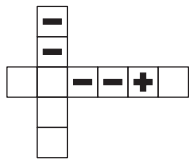
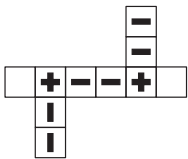


What is the sum of the three numbers on the vertices of the coloured triangle?

- (A) 12 (B) 14 (C) 15 (D) 17 (E) 18

SOLUTION: All numbers in the figure are divisible by 7, so it has to be in the middle. The left circle has to be 5, because only 105 and 210 are divisible by 5. To finish, we put 3 on the top, to make 105. Then we put 6 down, to make 210. The last one is 4, and goes to the right.

27 (SI). Which of the following nets cannot be folded into the solid 

- (A)  (B)  (C)  (D)  (E) 

SOLUTION: If you make a box of answer B, you have to flip the solid to make sure that the plus-sign is at the right side. If you do so with the the other possibilities, the minus-signs have the wrong direction on the top.

28 (UA). The four villages A, B, C and D lie along a road in that order. The distance between neighbouring villages is 10 km. There are 10 students who live in village A, 20 students who live in village B, 30 students who live in village C and 40 students who live in village D. The villagers want to build a school so that the total distance travelled by the students when going to school is as small as possible. Where should they build the school?

- (A) in A (B) in B (C) in the middle between B and C
(D) in C (E) in D

SOLUTION: School in D: $10 \times 30 + 20 \times 20 + 30 \times 10 = 1000$ km

School in C: $10 \times 20 + 20 \times 10 + 40 \times 10 = 800$ km

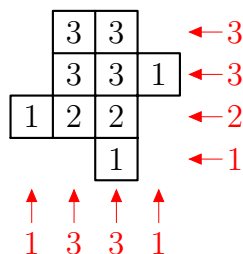
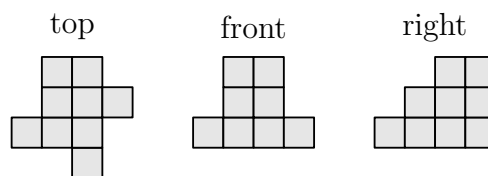
School between B and C: $10 \times 15 + 20 \times 5 + 30 \times 5 + 40 \times 15 = 1000$ km

School in B: $10 \times 10 + 30 \times 10 + 40 \times 20 = 1200$ km

School in A: $20 \times 10 + 30 \times 20 + 40 \times 30 = 2000$ km

29 (SK). The three pictures show a structure made from cubes as seen from the top, from the front and from the right. What is the maximum number of cubes that could have been used to build the structure?

- (A) 18 (B) 19 (C) 20 (D) 21 (E) 22



SOLUTION:

30 (HU). 30 people are sitting round a circular table. Some of them are wearing a hat. Those who do wear a hat always tell the truth while those who do not wear a hat can either lie or tell the truth. Each person says "At least one of my two neighbours is not wearing a hat." What is the largest number of people who could be wearing a hat?

- (A) 5 (B) 10 (C) 15 (D) 20 (E) 25

SOLUTION: At least one of three neighboring persons could not be wearing a hat, so there are at least 10 persons who do not wear a hat. The following arrangement is possible: NHHNHHNHH... (N: no hat. H: hat) Hence at most 20 could be wearing a hat.